Gukura: Child Stunting in Rwanda, District by District

NISR Big Data Hackathon 2026 · Track 3: Open Innovation (Health)

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- **Live app:** https://pacyuzu16.github.io/NISR/
- **Source code:** https://github.com/pacyuzu16/NISR
- **Data:** Rwanda Demographic and Health Survey 2025 (NISR, MOH, ICF): Final Report and microdata

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1. The problem

Stunting means a child is too short for their age. It is the clearest sign of long-term undernutrition and poor care in early life, and its effects on health, learning and earnings last into adulthood.

- In 2025, **26.8%** of Rwandan children under five were stunted (95% CI 25.4–28.1%), down from 51% in 2005.
- The second **National Strategy for Transformation (NST2, 2024–2029)** makes reducing stunting a headline priority, with a target of **below 15% by 2029**. That leaves almost 12 percentage points to cut in four years.
- The national average hides large gaps. District stunting ranges from **8.7% (Nyarugenge) to 38.8% (Gicumbi)**, and it is **40%** among children in the poorest households against **9%** in the richest.

District planners, the National Child Development Agency (NCDA) and partners need to know **where** to act first, **which** household conditions matter most, and **how much** improvement to expect. The published report answers these questions in hundreds of pages of tables. Gukura turns them into a tool.

2. The solution

Gukura ("to grow" in Kinyarwanda) is a web app with five pages.

Page	What it answers	Main features
Overview	How big is the problem, and how close is the NST2 target?	Headline estimate with confidence interval, trend since 2005 against the 2029 target, stunting by age in months
Districts	Where is stunting highest, and where do most stunted children live?	Map of all 30 districts, four indicators, ranked list with 95% CIs, search and province filter, district profiles, priority level

Page	What it answers	Main features
Drivers	Which factors go with stunting once everything is weighed together?	Raw gaps by 13 characteristics with CIs; chart of adjusted odds ratios with plain-language interpretation
Scenarios	What would it take to reach the NST2 target?	Coverage sliders for seven programmes and presets; projected rate for Rwanda or any district against the 15% target; share of the gap closed; before/after for all 30 districts
Risk check	For a given child, what is the estimated risk, and what would lower it?	Profile form with validation and examples; estimated risk against the national rate; effect of each modifiable improvement
Methods	Can the numbers be trusted?	Data sources, methods, AUC and calibration chart, limitations, privacy

What makes it useful

1. **Rate *and* burden.** The *share of stunted children* view shows that **Rubavu (6.0%) and Nyagatare (5.9%)** hold the largest numbers of stunted children in Rwanda, even though neither has the highest rate. Planning on rates alone would miss them. A *rate vs. number* chart places every district on both measures and highlights the **12 districts that are above the national rate and hold more than an equal share** of stunted children.
2. **Uncertainty shown everywhere.** Every district estimate has a 95% confidence interval. The district profile says whether a district is *clearly* above or below the national figure, or not clearly different.
3. **From evidence to action.** Each district profile states in plain words where the district is behind Rwanda (poverty, mothers' schooling, birth spacing, toilets, insurance) and links straight to the risk check for that province. **Print brief** turns any profile into a one-page A4 summary for district officials and Joint Action Development Forum meetings, and the full district table downloads as CSV.
4. **Peer districts.** Each profile lists the districts whose households look most alike but have lower stunting, so planners know where to look for practices that work.
5. **Policy simulation.** The Scenarios page shows how far each programme would move stunting, and how close Rwanda would get to the NST2 target.
6. **Honest about the model.** Effects that aren't statistically clear are shown in grey and flagged in the risk check.

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3. Data

Source	Use
RDHS 2025 microdata: household-member recode (PR) and children's recode (KR), from the NISR Microdata Catalogue (public-use terms)	All stunting estimates, confidence intervals, the model and district context
RDHS 2025 Final Report, Table D.4	Severe stunting, wasting, underweight and overweight by district
Earlier RDHS reports (2005–2019/20)	National trend

Source	Use
NST2 (2024–2029)	Target of below 15% stunting by 2029
geoBoundaries, Rwanda ADM2 (CC BY 4.0)	District boundaries for the map

Definition: a child is stunted when their height-for-age z-score is below −2 SD of the WHO Child Growth Standards median.

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4. Methodology

4.1 Estimates

- **Population:** de facto children under five with a valid height-for-age measurement (n = 7,261), as in the official report.
- **Weighting:** the survey's household sampling weights.
- **Uncertainty:** 95% confidence intervals by Taylor linearisation over the two-stage cluster design (sampling units within strata).
- **Validation:** our national (26.8%) and all 30 district estimates reproduce Table D.4 of the Final Report exactly. The build script fails if any district differs.
- **Share of stunted children:** the weighted number of stunted children in a district divided by the national weighted total.

4.2 Model

- **Outcome:** stunted (yes/no) for 6,715 living children of interviewed mothers (children's recode).
- **Predictors (13):** child's age, sex and size at birth; recent diarrhoea; mother's education, height, health insurance, and the time since her previous birth; household wealth, drinking-water source and toilet type; province; urban or rural residence.
- **Missing data:** mother's height (measured in a subsample) and size at birth (asked only for recent births) are kept as separate groups, so no child is dropped.
- **Estimation:** survey-weighted logistic regression with cluster-robust standard errors (statsmodels).
- **Model choice:** we compared it with gradient-boosted trees using 5-fold stratified cross-validation. The logistic model was at least as accurate and is far easier to interpret, so we kept it.

Model	Cross-validated AUC
Weighted logistic regression (chosen)	**0.708**
Gradient-boosted trees	0.693

Calibration is good: across tenths of predicted risk, observed stunting closely tracks the prediction (chart on the Methods page).

4.3 Main findings (adjusted odds ratios, 95% CI)

Factor	Group vs reference	Odds ratio
Household wealth	Poorest vs richest	**4.64** (3.39–6.35)
Mother's height	145–149 cm vs 155 cm or more	**2.96** (2.19–4.00)
Size at birth	Smaller than average vs average or larger	**2.23** (1.86–2.69)
Mother's education	No schooling vs higher	**2.43** (1.36–4.36)
Child's age	12–23 months vs 0–5 months	**2.38** (1.83–3.08), the peak risk window
Birth interval	Under 2 years vs 4 years or more	**1.41** (1.09–1.83)
Toilet type	Unimproved vs improved	0.99 (0.84–1.16), no independent effect

What this means: 1. Poverty is the biggest single factor. 2. Stunting passes between generations, through the mother's height and the baby's size at birth. 3. **Birth spacing is the strongest factor families can change.** 4. Sanitation alone doesn't separate stunted from non-stunted children once wealth is accounted for. That doesn't make it unimportant.

4.4 Scenario simulation and peer districts (machine learning for planning)

- **Scenarios (g-computation).** For each of seven programmes, we predict every surveyed child's stunting risk with the programme applied, using the fitted model, and average with survey weights for Rwanda and each district. The programmes are birth spacing (4+ years), health insurance, diarrhoea prevention, safe water, sanitation, social protection (the poorest 40% move up one wealth level) and mothers' schooling (one more level). All 2^7 = 128 combinations are precomputed. With independent coverage p for each programme, the expected rate is $\Sigma_S \Pi_{i\in S} p_i \Pi_{i\notin S} (1 - p_i) \cdot \text{risk}(S)$, so any slider setting is exact in the browser and no child-level data is shipped. The projected rate is the official estimate plus the model-predicted change.
- **Peer districts (nearest neighbours).** Districts are compared on five standardised household measures. For each district we list the closest matches with stunting at least 3 points lower.

What the simulation shows:

Programme at full coverage	Change in national stunting
Mothers' schooling (one more level)	−2.1 points
Birth spacing (4+ years)	−1.9 points
Social protection (poorest 40% up one level)	−1.8 points
Health insurance	−0.4 points
Diarrhoea prevention	−0.4 points
Safe water	−0.3 points
Sanitation	no change
All seven together	**−6.5 points (26.8% → 20.3%)**

Even all seven programmes at full coverage close only **about 55% of the gap** to the NST2 target of 15%. Reaching the target will also need what the survey model can't capture, chiefly children's diets and feeding practices. That is a clear, evidence-based message for planners: no single programme will get Rwanda there.

5. Technology

Layer	Choice	Why
Analysis	Python: pandas, statsmodels, scikit-learn	Survey-weighted estimation and model comparison
Output	One aggregated JSON file (about 30 KB)	No respondent-level data ever reaches the app
Front end	React + TypeScript + Vite	Fast and typed, with no server to maintain
Charts and map	Custom SVG components, d3-geo	Full control of accessibility, theming and confidence intervals
Hosting	GitHub Pages via GitHub Actions	Free, and deploys automatically on every push

Usability and design: - Works on phones (bottom tab bar, pinned risk result), tablets and desktops. - Light and dark themes, using colour palettes checked for colour-blind readers. - Full keyboard navigation with visible focus, a table view for every chart, and screen-reader labels. - **Automated accessibility audit (axe-core): 0 violations** on all five pages, in light and dark themes, at desktop and phone widths. - Loading, empty, error-with-retry and form-validation states. - Links can be shared, e.g. `#/districts/Gicumbi`, and show a preview card when shared on WhatsApp or by email.

6. Impact

- **District planners and Joint Action Development Forums:** see where their district stands, how sure the estimate is, and which household factors it is behind on.
- **NCDA and the Ministry of Health:** target programmes by burden (number of children) as well as by rate.
- **Community health workers and partners:** use the risk check in planning conversations to show how birth spacing, insurance and illness prevention change a child's chances.
- **Planning:** the Scenarios page lets planners test combinations of programmes for their district before committing budgets, and shows which combinations would bring a district below 15%.
- **Accountability:** progress against the NST2 target is visible at a glance, and the tool can be updated when the next RDHS round is released by re-running one script.

7. Limitations

- The survey is a snapshot at one point in time, so the model shows associations, not causes.
- District samples are small (about 170–550 children each), which is why every district estimate shows a confidence interval.

- Diet and feeding practices aren't in the model yet. They are a natural next step.
- The risk check is for planning conversations, not for screening individual children.

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8. Privacy and data ethics

The RDHS microdata was obtained under NISR's public-use terms and is never published. It is excluded from the code repository, and the app contains only aggregated estimates and model coefficients. No attempt is made to identify respondents.

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9. How to run

```
# Rebuild the data (requires the RDHS 2025 microdata in hackathon/microdata/)
cd hackathon
python3 -m venv .venv && .venv/bin/pip install -r analysis/requirements.txt
.venv/bin/python analysis/build_outputs.py

# Run the web app
cd web && npm install && npm run dev
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10. Team and contributions

Gukura was conceptualized, designed, and developed by **Team NavySec**: - **ISHIMWE Marie Pauline** (Team Leader, Computer Engineering, UR-CST): nutrition indicator research, policy target alignment (NST2), and domain validation against official RDHS 2025 reports. - **CYUZUZO Pacifique** (Computer Engineering, UR-CST): survey-weighted statistical analysis pipeline (Python/statsmodels), interactive web application development (React, TypeScript, custom SVG visualizations), and deployment.

We used an AI assistant for limited support with code and drafting. The team designed, reviewed and is responsible for all of the work.

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11. References

- National Institute of Statistics of Rwanda (NISR), Ministry of Health (MOH) and ICF (2026). *Rwanda Demographic and Health Survey 2025: Final Report.* Kigali and Rockville, MD.
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